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Simulation and Gaming

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Farmer Preferences on Contested Land Use in Agroforestry Landscapes: The RESORTES Game Implementation in Indonesia

Rika Ratna Sari[1], Danny Dwi Saputra[1], Rizki Maulana Ishaq[1], Eka Purnamasari[1], and Erika N. Speelman[2]

Abstract

Background. Achieving sustainable landscape management in complex social-ecological systems necessitates a holistic evaluation and a participatory approach. Agroforestry is broadly believed to be a sustainable land-use practice. However, the existence of agroforestry systems remains uncertain, as it is influenced by socio-economic factors that depend on farmers as landowners.

Objectives. This study evaluated (1) chosen land use (LU) systems by agroforestry and crop farmers, (2) the consistency of LU choice, and (3) social factors influencing the relationship and leader role on the farmer decision-making process using a serious game.

Methods. We adapted the RESORTES board game to test applicability to the Indonesian landscape and implement the game as a research tool in the Malang context, Indonesia. Three game sessions were conducted involving the representatives of agroforestry and crop farmers of various ages, including the societal

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leader figure in two villages. All participants choices and (social) interactions during the session were recorded and analysed qualitatively and quantitatively.

Results. The RESORTES game was played appropriately in all sessions with a minor adjustment to the game, attributed to better matching the local relevance. The result showed that farmers' preference for LU choices represented their real-life practices. Approximately 70% of plots selected by farmers in both villages were tree-based LU (agroforestry), with relatively high consistency across game rounds. Younger farmers tended to choose crop-based LU due to more economic benefits provided by the system. Older farmers were more firm on tree-based (agroforestry) systems to fulfil their daily needs; they also benefitted from less labour and the low operational costs needed by agroforestry systems.

Discussion. The relations (relatedness index) among farmers and the role of the leader (respected figure) became important factors in building good communications and trust, which influenced farmers' decision-making on collaborating to maintain or change their LU. These key factors could be useful for community-based LU planning in the future, which takes into account local preferences.

Keywords
tree-based system, participatory approach, board game, agroforestry

Introduction

Managing the trade-off between economic and ecological considerations in accomplishing numerous Sustainable Development Goals (SDGs) in the agriculture sector is challenging, especially in landscapes where there is a connection between forested areas and cleared-field patches that collectively offer a range of benefits to different stakeholders (Jackson et al., 2007; Speelman et al., 2006). The deterioration of ecosystem services is caused by unsustainable land-use (LU) change, which is driven by institutional, market, and regulatory aspects (Ribeiro Palacios et al., 2013). As a result, the development of more environmentally friendly agricultural landscapes has become increasingly important for many institutions and organisations (Astier et al., 2012; Wegner & Pascual, 2011). Designing a sustainable agricultural landscape requires the participation of local stakeholders, as part of non-forestry land use (local: *area penggunaan lain*-APL) is privately managed by smallholder farmers. It is important to understand the local preferences and the decision-making processes behind adopting specific LU types.

In the present day, farmers' decision-making is often impacted by frequently conflicting schemes (Speelman et al., 2014). It is influenced by a broad range of elements involving individual decisions, communal (social) pressures, and their socio-demographic background (age and education) (Bui & Nguyen, 2021; Duong et al., 2019; Zhumanova et al., 2016). The implementation of economic incentives such as

payments for environmental services to promote sustainable agriculture practices can lead to competition and individualism among community members. However, it could facilitate cooperation among smallholder farmers to acquire significant environmental services, mostly obtained beyond individual farms. The selection of land management practices by farmers is primarily influenced by several aspects, including economic benefit-cost analysis, ecological advantages, resource availability, market demand, and market access (Bui & Nguyen, 2021; Sari et al., 2024). Since private areas, such as those in Indonesia, are fully controlled by farmers, the probability of LU conversion is higher compared to the area belonging to the government. Apart from several factors aforementioned, which are mostly under the domain of instrumental value, relational value (harmony-oriented)- which closely relates to the social-cultural aspect embedded in the community- also plays an important role (Githinji et al., 2023; van Noordwijk et al., 2023). The rationale for this decision-making process must be comprehended since it is crucial to be involved in formulating a strategy to accomplish sustainable agriculture management, particularly on privately owned land.

In hilly mountainous landscapes, when tree- and crop-based LU are complementing each other and are combined into agroforestry systems, the risk of changes in the LU system may be more severe. The decrease in agricultural productivity is not just attributed to the degradation of soil quality resulting from ongoing erosion in mountainous terrain (Andriyanto et al., 2015; Rijsdijk et al., 2007) but also to catastrophic events like volcanic eruptions (Maeno et al., 2019). The 2014 eruption of Mt Kelud in East Java, Indonesia, caused disruptions to the crop-based LU system in the Kali Konto sub-watershed. This event resulted in changes in soil physical characteristics, including a significant decline in soil infiltration rate (Saputra et al., 2022) and the development of a hydrophobicity layer due to the deposition of volcanic ash (Saputra et al., 2023), leading to a decline in coffee bean production (Ishaq et al., 2020). Due to climate change and the COVID-19 pandemic, it is expected that smallholder vulnerability will increase, leading farmers to either intensify their existing practices or alter their current LU system. Currently, the research in this field has been centred on (social-)ecological investigations aimed at comprehending the effectiveness of different LU systems in conserving ecosystem services (Saputra et al., 2023; Sari, Rossyda, et al., 2023). There is limited evidence available on farmer preferences on LU system options that are investigated using participatory methods, focusing on how farmers select their LU systems, including the influence of social aspects on the decision-making process from the perspective of the social-ecological context.

Acquiring the preferences of smallholder farmers and comprehending the main factors influencing their decision-making process is challenging, especially when using conventional methods like direct interviews (Sari et al., 2024). The quality of the data obtained through interview methods might rely on the farmer's attitude and the skill (flexibility and adaptability) of the facilitator in extracting information (Azadegan & Kolfschoten, 2014). Moreover, conventional methods sometimes direct the participant's answer in certain ways to pre-prepared questions. Another novel technique that might be explored is employing a serious game that emphasises a participatory

methodology. Nowadays, serious games have been designed as tools for debate and decision support tools in the fields of agriculture and natural resource management (ComMod, 2013) and have been used for various purposes for research (Sari et al., 2024) and intervention (Rodela et al., 2019), including increasing public knowledge of certain topics like water, sustainability, and social difficulties to stimulate inclusive business (Speelman et al., 2019) as well as the exploration of possibilities in the socio-environmental-political domain (Damani et al., 2015; Harteveld et al., 2007; Orduña Alegría et al., 2020). This approach enables players to effectively simulate and examine the advantages and difficulties of intricate communal LU decision-making while engaging in a learning process, as demonstrated in the design of the RESORTES board game (Speelman et al., 2014). The RESORTES is a closed and realistic game that involves LU decision-making, portraying an agricultural landscape and including several of the current challenges faced in complex agricultural landscapes.

In order to gain a holistic social-ecological understanding, it is necessary to comprehend farmers' preferences and decision-making processes using participatory approaches. This information will complement the existing study conducted earlier, which focused only on ecological research. In this paper, we aimed to use the RESORTES game as a research tool to understand how smallholder farmers make decisions on LU selection in their landscapes. However, we firstly tested the adaptation of the RESORTES game to be used in agroforestry landscapes in the Indonesian context. We developed five research questions, where the first two questions focused on testing the performance of the RESORTES game and the last three questions focused on understanding the farmer decision-making process. Our research questions were (1) Can the RESORTES game be effectively implemented in Indonesian mountainous landscapes? (2) How do LU choices in the game relate to those in real life? (3) How are farmer decisions distributed in selecting agricultural plots within the landscape? (4) How does farmer LU's decision vary and remain consistent across different types and ages? (5) How do social relations (i.e., communication and leadership) influence the decision-making process? The findings of this study can offer valuable insights to local authorities regarding farmer preferences and decision-making processes within complex social-ecological systems. These insights can then be considered when planning future LU, aiming for more sustainable landscape management.

Materials and Methods

Study Location

The study was conducted in the Kali Konto sub-watershed, located in East Java, Indonesia. Specifically, the research focused on the Tulungrejo and Sumberagung villages in Malang Regency (Figure 1). The role of this landscape for the Brantas basin is crucial for sustaining the livelihoods of almost half of the inhabitants in the East Java region. The agroforestry landscapes in this area depict a mosaic of trees and crops, with coffee (*Coffea canephora*) as the predominant cash crop, accompanied by fruit (*Durio*

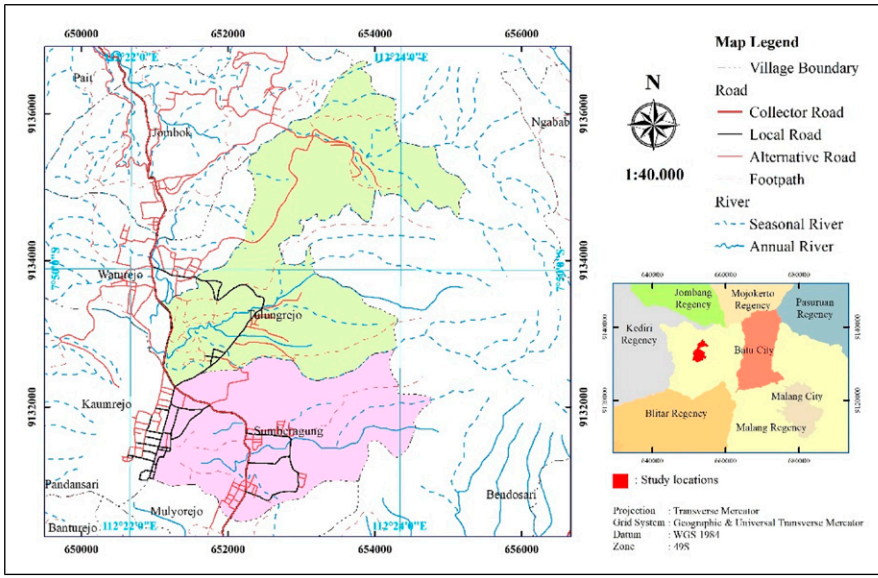


Figure 1. Study location: Tulungrejo and Sumber Agung villages, Malang Regency, Indonesia.

zibethinus and *Parsea americana*, *Artocarpus heterophyllus*, and *Nephelium spp.*), legumes (*Molucana falcataria*, *Gliricidia sepium*, *Leucaena leucocephala*, and *Parkia speciosa*), and timber (*Tectona grandis*, *Swietenia mahagoni*, and *Anthocephalus cadamba*). Maize, Napier grass, and cabbage were the predominant annual crops (Sari, Rossyda, et al., 2023). The research area exhibits a tropical monsoon climate and is characterised by hilly terrains with steep inclines, which serve as water towers. The annual precipitation ranges from around 2995 to 4422 mm. The extent of natural forest cover is diminishing, with approximately 20% of the original forested area remaining. According to Dewi et al. (2017), the population density is around 453 individuals per km^2 (van Noordwijk et al., 2020). The majority of the population consists of indigenous people, specifically the Javanese ethnicity. Additionally, 11% of the population lives below the national poverty level. The agricultural focus of smallholder farmers in the study area varies from subsistence farming to market-oriented farming (Sari et al., 2024).

Research Steps

The study was carried out in six stages, as illustrated in Figure 2. Initially, we collected secondary data from existing literature on the history of agricultural plots and the educational background of smallholder farmers (BPS, 2022). This data included information on socio-economic factors such as education, poverty levels, and the total size of the population of farmers in each village. Using that information, Tulungrejo and

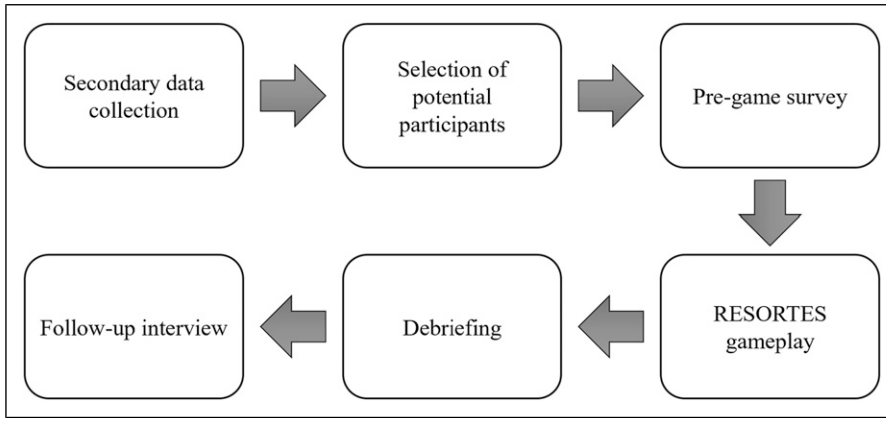


Figure 2. The six research steps required to performed the RESORTES games in the study location.

Sumberagung villages were selected because the population of farmers reached more than 50% of the total (62% and 52%, respectively) based on the national database (Central-Bureau-of-Statistics, 2016). We chose farmers who have represented the prevalence of tree-based, crop-based, and mixed farmers to obtain different perspectives on LU choices, as these LUs dominate the landscape. To examine whether social dynamics influence LU selection, we intentionally included farmers with varied roles in the community in each session, differentiating between ordinary farmers and respected group leaders, considering their hierarchical relationships.

We employed a purposive sampling method to select a total of 18 smallholder farmer representatives based on the aforementioned criteria (Table 1); however, only 83% of selected farmers attended (Appendix 1). Forty percent of the participants were young farmers who were under 50 years old, while the rest were older farmers (>50 years old) with education levels ranging from elementary to junior high school (Table 1). To ensure the diversity of participants and to minimise a selection bias in the project, we 1) used a neutral invitation message that emphasised participation in simulating daily farming practices without indicating the study's focus; 2) employed diverse recruitment channels, recruiting participants from multiple sources such as farmer associations, village chiefs, and extension officers; and 3) implemented balanced participant selection by applying predefined inclusive criteria as mentioned in Table 1. We divided the participants into three groups (sessions), each of which played the RESORTES game separately. Within each group, six players were selected based on their divergent preferences in LU systems to facilitate more substantial communication and discussion throughout the implementation of LU planning during gameplay. As our study sought to examine farmers' decision-making processes regarding land use and sustainability, participant selection prioritised economic roles and farming practices rather than

Table 1. Characteristics of Participants.

Criteria	Group 1	Group 2	Group 3
Location	Sumber Agung	Tulungrejo	Tulungrejo
Farmer type (domination)	Agroforestry farmer	Crop-based farmer	Agroforestry and crop-based farmer
Societal role involvement	1 social leader (local: kamituo)	2 social leaders (local: kamituo)	1 group leader
Age range (years)	30–60 (mean: 51)	36–65 (mean: 55)	34–58 (mean: 45)
Cultivation area (ha)	0.5–1 (mean: 0.8)	0.25–1 (mean 0.7)	0.25–1 (mean: 0.5)
Education	Junior high school	Elementary school	Elementary school
Poverty level (%)*	11	11	11

*Note: Based on percentage of population below the national poverty line.

gender, as decision-making in this context is typically collective, involving both men and women.

A pre-game survey was undertaken by conducting in-depth interviews to collect information about present agricultural practices (LU systems), planned LU systems, the relationship between selected players, and their background (age, education level, and role in the community). The relationships between participants were categorised as family, friendship, and/or religion-related ties. Three game sessions were organised, each comprising six farmers in separate groups. Each session represented the dominance of tree-based, crop-based, and mixed farmers. The implementation of the RESORTES game typically lasted around 60 to 90 minutes per session, during which a maximum of 5 rounds was played. After the games ended, participants engaged in a debriefing session to analyse, clarify, and discuss the strategies implemented during gameplay. These sessions focused on exploring farmers' aspirations in land-use choices through 'why' questions and sharing between participants. Subsequently, follow-up interviews were conducted with all participants 2-3 days after the game. These interviews aimed to collect information about the purpose and advantages of their participation in the game.

Prior to participation, all participants were informed about the study procedures and their rights, including voluntary participation and the option to withdraw at any time. Informed consent was obtained from all participants before data collection. The study also ensured the confidentiality and anonymity of all responses, as it was approved by the Institutional Review Board (IRB).

RESORTES Game Description

We adapted the RESORTES board game, designed by Speelman et al. (2014). RESORTES is a participatory game that engages local communities and farmers in the process of LU planning in agricultural landscapes. This game facilitates the exploration

of farmers' preferences and enables the identification of patterns and important aspects that primarily influence farmers' decisions in managing and deciding their LU system. This game involves two important factors: (1) the risks and uncertainties associated with financial returns from the LU system and (2) the coordination of LU activities among farmers to maximise financial benefits from environmental service payment schemes such as payment of environmental services (PES) and economies of scale (EOS). Within this context, we considered only involving PES due to its relevance in the local setting.

The RESORTES's game board consisted of 37 interconnected hexagons, partitioned into four equal-sized quadrants of 9 hexagons each to depict the hilly terrain (Figure 3) and a central hexagon representing the village. The game elements included 4 (four) sets of LU cards, each set containing 24 cards. These cards represented various types of LU systems with differing levels of risk. Additionally, there were 24 land plot cards customised based on the preferences of the farmers surveyed before the game. The equipment also included two dice with different number ranges, a scoring board, and dummy money. We implemented the game mechanics and rules based on Speelman et al., 2014. However, we modified the LU options, specifically using tree-based and crop-based systems instead of forest-based and cleared-field options as mentioned in the original version. This adjustment was made to better match the agroforest landscapes in Malang and increase the engagement by imitating the local environment.

Prior to starting the game, the participants were provided with information regarding the arrangement and the game rules, which included (1) they would be colonising a territory with four potential LU types, assigning one LU to each field; (2) each LU

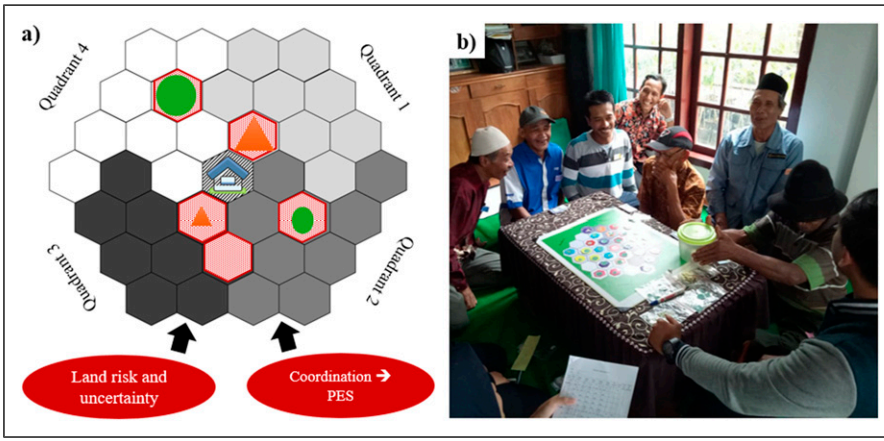


Figure 3. a) RESORTES's board game with plots located in four quadrants, with residential plots situated in the middle. Differences in plot choices and LU systems are shown in different shapes (circles and triangles). Various sizes indicate the level of land risk (low and high) (modified from Speelman et al. (2014)); b) Game session in Tulungrejo villages, where the participants were predominantly tree-based farmers.

would result in varying levels of benefits depending on the risk due to fluctuations in price or production; (3) coordinating with other participants to generate additional benefit (points); and (4) participants with the highest number of points become the winner. To start the session, preliminary trial rounds were conducted to ensure all participants fully understood the rules and gameplay mechanics. Once familiarised, players take turns selecting field locations and subsequently assign land-use types to their respective fields. Following the finalisation of these selections, dice were rolled to determine additional points based on planning schemes, after which scores were calculated. The game continues for around 4-5 rounds as participants choose. To align with local conditions, several adjustments might be made without altering its original objective and settings (including variables).

Data Monitoring

To examine our research questions, we carefully recorded and observed several factors throughout the games, arranging the data in a tabular form for each session. We conducted this monitoring by splitting it into two categories according to our research objectives, namely (1) testing the game performance and (2) understanding the game outcome: the preferences of farmers for LU system choices and their degree of consistency.

Regarding the game's outcome, we monitored the LU system chosen by each player in every round and session to assess the degree of consistency in their decision-making. We tracked all communication between each participant during the session, which influenced the alteration of the LU decision. We recorded all interactions made by participants, including the number of relevant comments made by participants and the points earned based on their individual decisions and collaboration with others.

The game performance was monitored by recording the active participation during the session. The level of active participation was determined by the engagement and involvement of individual participants throughout the gameplay. Active participants were defined as those who often contributed input and relevant comments during the process of game implementation, discussion groups, and follow-up interviews conducted after the game (post-game survey). The quantity of comments and feedback from each participant was counted in each round.

We used both direct observation and audio recording to cross-check the observation data. A total of two assistant facilitators (in addition to the game master/main facilitator) were needed: one responsible for recording all participant choices and points earned throughout the session to ensure accurate data collection and the other for documenting social interactions, including the number of relevant comments and participant dynamics. The qualitative analysis of the data from group discussions and follow-up interviews focused on players' experiences during the game (Patton, 2001). To verify if the choices made by participants during gameplay accurately reflected their real-life decision-making, we conducted a comparison between each participant's LU system selections and the data we collected from the pre-game survey.

Data Analysis

The data collected, both qualitatively and quantitatively, during the gameplay was organised into an Excel spreadsheet. We employed network graphs to visually display the social connections within each group of participants. The relatedness index per player was determined as the normalised node strength, as defined by Boccaletti et al. (2006). This index is based on the number of high-valued ties, such as family, friendship, and religion-related connections.

We calculated the percentage of participants who selected tree-based land use by dividing the number of times a tree was selected in each round by the total number of game rounds, similar to the calculation for high-risk LUS selection. Consistency was calculated by determining how many times a farmer maintained the same choice in each round, divided by the total number of game rounds. To examine the variations in LU system choices and the consistency of change among different farmer types and age groups, we used a two-way analysis of variance (ANOVA). We performed ANOVA type-III sums of squares as our data was unbalanced due to the absence of a few invited participants. We conducted a correlation analysis to examine the relationship between the number of comments, participants' ages, relatedness index, additional points obtained via coordination, and total points earned by each participant. The statistical analyses were performed using R 4.2.0 (R-Core-Team, 2022).

Results

Implementation of RESORTES Game in the Context of Study Location

We conducted the RESORTES game in three separate groups (game sessions) in two villages (Sumberagung and Tulungrejo), with various farmer types. The RESORTES game was played proficiently in all sessions, as evidenced by the participants' enthusiasm during playtime, the farmer's eagerness to complete the game until the final round, and the constructive outcome that resulted from each session. Through our careful observation in each session, we were able to identify some dynamics that emerged throughout games. The participants from group 1 achieved the highest total points, averaging 126 points compared to groups 2 and 3 (averages of 119 and 81 points, respectively). The high scores received by group 1 were a result of additional points earned by the collaboration among participants to attain a state where the environmental services function incentive (PES) was fulfilled by creating 8 tree-based plots out of the total of 9 plots in each quadrant. The extra points acquired as a result of this collaboration averaged 27 points per participant (Appendix 4). Those results imply that the RESORTES game setting allows participants to interact and engage in open and dynamic discussions.

In the context of participants' choices during the game session, we compared participants' LU choices during the game sessions and actual practices. We found that approximately 86% of participants made the same LU choices as their actual

LU when playing the game. Agroforestry farmers tended to choose tree-based systems during the game sessions. It indicates that the decision-making during the game session was made based on their real-life experience. This result is supported by our post-game survey, where the majority of participants (88%) acknowledged that the selection of LU systems in the game was largely influenced by their actual practices. While the rest of the participants mentioned that the decision-making was based on their preference, which has not been implemented in real-life situations.

Preferred Agricultural Plots Distribution in Landscape

Based on the game results, we have discovered an interesting pattern in how agricultural plots were chosen and spatially distributed in the landscape during game sessions. The plots for group one, which was dominated by agroforestry farmers (Sumberagung village), were evenly distributed over the three or four quadrants, as shown in Figure 4. On the other hand, the other two groups from Tulungrejo village chose their plots in a centralised manner, limiting their selection to only one or two quadrants. The majority of farmers (agroforestry and crop farmers) often choose the initial plots located in close proximity to the settlement area owing to the convenience of access. Farmers that favoured centralised plots prioritised efficiency in maintenance due to labour and fertiliser transport. Conversely, farmers who choose to distribute their plots aimed to address two main issues: firstly, the variance in soil fertility at different elevations in the landscape, and secondly, the need to reduce the risk of crop failure caused by pest infestations.



Figure 4. Spatial distribution of the farmer's plot occupation in the agricultural landscape with the settlement area located in the center. Each colour symbolised a distinct ownership. a) Fully distributed plot occupation (group 1, dominated by agroforestry farmers at Sumberagung village). b) Centralised plot distribution (groups 2 and 3, dominated by crop farmers at Tulungrejo village).

Farmer Preference on Land-Use Choices Across Different Types and Ages

To examine the differences in participants' responses regarding their LU selection and the consistency of their decisions, we conducted a statistical analysis, as shown in Table 2. Our analysis revealed substantial variations in farmer preferences during the game sessions across farmer types and ages (Table 2). In all groups, the majority of participants selected a high-risk land-use system (Appendix 3). However, we were unable to uncover any conclusive data indicating that the participants' choice of LU system posed risks in their decision-making during our gaming session (Table 2).

We found a significant proportion of agroforestry farmers that selected tree-based LU systems in Figure 5(a), as well as the high percentage of crop farmers choosing crop-based LU systems, as depicted in Figure 5(b). We observed that young farmers in Tulungrejo village exhibit a preference for crop-based LU schemes over tree-based ones. Overall, approximately 70% of plots selected by farmers in both villages were tree-based systems (Appendix 2). These choices were significantly different between farmer types, which indicates that during the session, the decision was made based on their actual experience.

Interestingly, between rounds 3 to 5, a noticeable shift in decision-making patterns emerged, with 60% of participants starting to imitate the choices of those who had accumulated the highest points. As the game progressed, farmers began to observe and

Table 2. Results of Unbalanced Analysis of Variance Examining Participants' Decision on LU Selection During RESOTES Game Session.

Response Variables	Source of Variation	df	*p* value
Selected tree-based LU	Location	1	0.260
	Farmer type	1	**0.003****
	Age	1	**0.027***
Selected high-risk LUS	Location	1	0.754
	Farmer type	1	0.903
	Age	1	0.824
Consistency of LU	Location	1	**0.011***
	Farmer type	1	0.405
	Age	1	**0.027***
Consistency of commodity	Location	1	0.744
	Farmer type	1	0.193
	Age	1	0.573
Number of comments	Location	1	0.077
	Farmer type	1	0.168
	Age	1	0.166
	Social role	1	**0.009****
Relatedness index	Location	1	**0.009****
	Farmer type	1	**0.001****
	Age	1	**0.001****

Note: df (degrees of freedom); Significant *p* values are displayed in bold**$p < 0.01$, *$p < 0.05$.

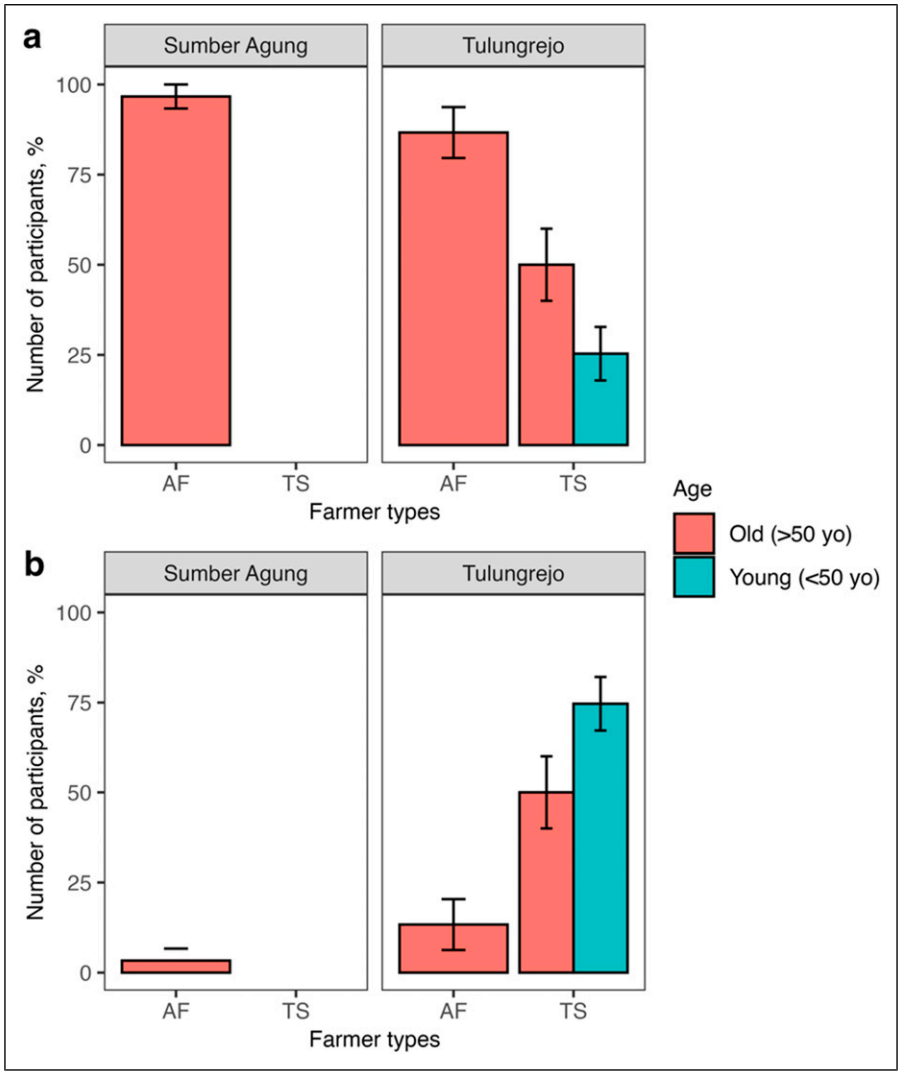


Figure 5. The variation of a) tree-based and b) crop-based land-use system (LUS) choices preferred by participants across different farmer type and ages (AF = agroforestry farmer; TS = crop farmers; Old above 50 years old; Young = below 50 years old).

adopt the strategies of top-performing participants, recognising their success in earning higher rewards. This trend was consistent across game sessions, indicating that social learning played a significant role in shaping decision-making behavior. However, we did not observe distinct differences in imitation patterns across farmer types or age groups.

Consistency of Agroforestry Farmers on Land-Use System Choices

Regarding the consistency of farmer decision-making and shifting LU systems during the game sessions, we observed a notable disparity across locations (groups) and farmer ages (Table 2). The level of consistency in decisions within group one (represented by agroforestry farmers in Sumberagung) was comparatively greater than in the other groups (Tulungrejo). During the game session in Tulungrejo, we observed that younger farmers exhibited more consistency in maintaining their LU strategies over 5 (five) game rounds compared to older farmers (Figure 6). We did not find any differences in farmers' consistency in terms of commodity alteration across locations and farmer ages (Table 2).

Individual Relationships and the Role of Societal Leader on Active Participation

We found that the relationship index differed across location (group), farmer type, and age (Table 2). Individual relationships between participants in each group are presented in Figure 7. These relationships demonstrated the level of attachment and engagement, which might have impacted behaviour throughout the game's implementation. The members of group 1 (from Sumberagung village) had a strong bond as they were all

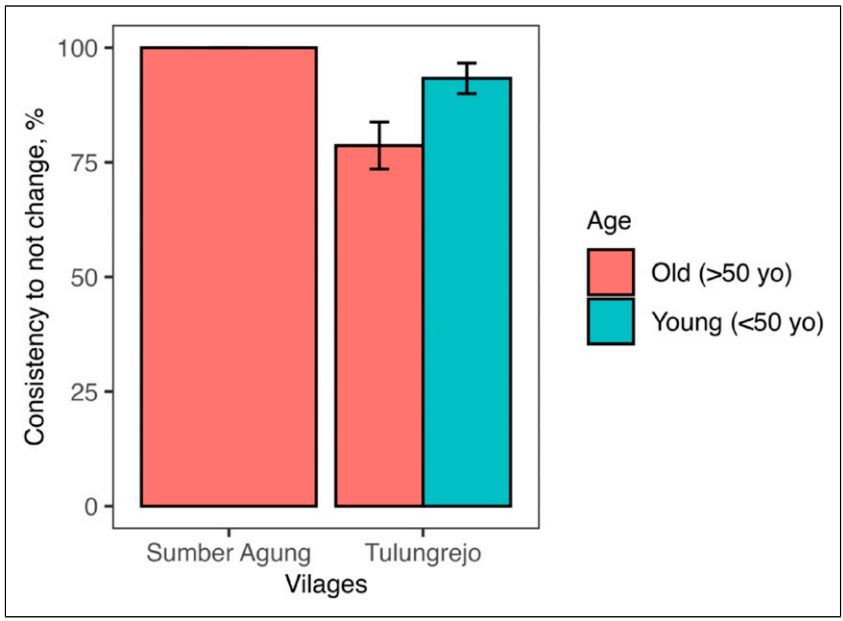


Figure 6. The degree of farmer's consistency to not change the land-use system (LUS) during the gameplay across locations and farmer age (yo=years old).

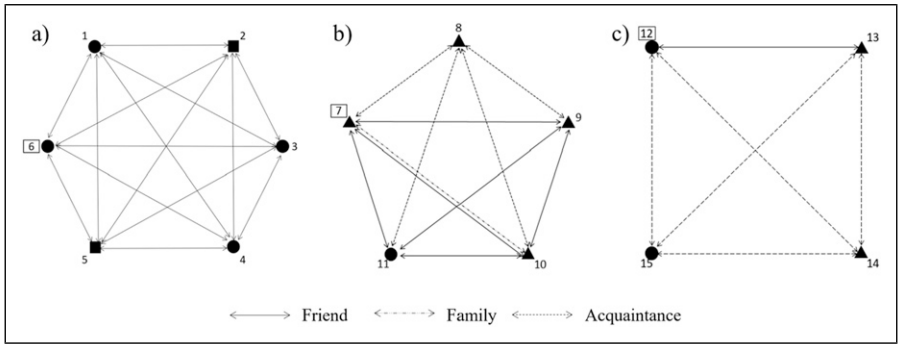


Figure 7. The relationships between participants in a) group 1—Sumberagung Village, b) group 2—Tulungrejo Village, and c) group 3—Tulungrejo Village. (Circle=tree-based farmers, triangle=crop-based farmers, square=farmers who manage both trees and crop-based systems. The respondent number in the box shows that the participant has a social role in the community).

familiar with each other. Each participant in this group identified the other five members as friends, as indicated by the direct connection line (Figure 7(a)). The participants in the second group (Tulungrejo Village) showed a similar pattern to group 1 but weaker. Several individuals were acquainted with each other, but their relationship could not yet be characterised as friendship (dashed line, Figure 7(b)). Two members of group two were related. In contrast, the interpersonal connections among individuals in group 3 were not profound. Participants in this group had limited communication with others, suggesting that they restricted the interactions to a small circle.

We found a positive and moderate correlation between the relatedness index and total comments in game sessions. It indicates that the degree of relationship within the group influences the active participation of farmers. Based on quantitative and qualitative analysis, the total comments (suggestions) in group 1 tended to have intensive communication, reaching 43 comments in total, with an average of 7-8 comments per person. In contrast, in other groups, there were only between 3-5 comments per person, with a low relation between participants (see also Figure 7(b) and (c)). We observed a positive association between the relatedness index and additional points generated from collaboration efforts (Figure 8(b)).

We observed that participants with societal leadership roles provided around twice as many suggestions compared to ordinary participants across groups/sessions (Figure 8(a)). Throughout the game, these participants actively participated in group conversations, offering comments and input to other participants. Approximately 40-50% of the overall remarks in each group were contributed by these participants. The extensive participation of the group leader seemed to have a substantial impact on individual farmers' decision-making. Nevertheless, farmers in Tulungrejo village were unlikely to consistently comply with the leader's suggestions (group 3) due to the somewhat distant relationship between the social leader and the farmers.

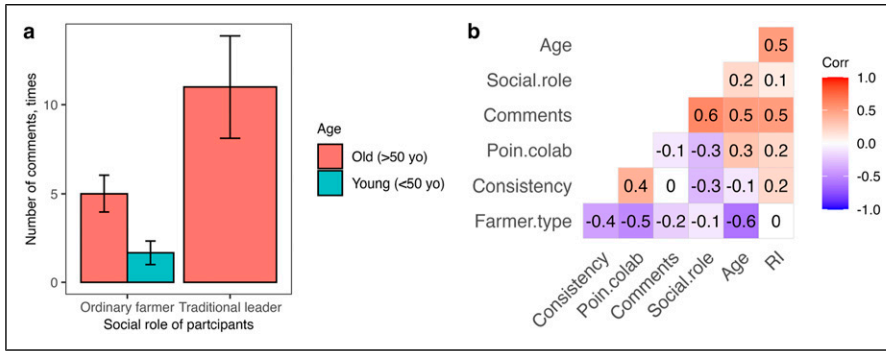


Figure 8. a) The role of social leader in providing input and suggestions during gameplay, b) Correlation between farmers backgrounds, social relationships, and the consistency of land-use system choices (Poin.colab= the points generated from collaborating with other participants during the game session; RI=relatedness index; yo=years old).

Lesson Learned Based on the Participants during the Gameplay

The gameplay of RESORTES provided valuable insights into farmer decision-making, particularly in terms of collaboration, perception of gaming as a learning tool, and risk-taking behaviour among different age groups. These observations highlight the potential of serious games as an engaging and insightful method to explore real-world agricultural challenges. One of the most significant lessons learned by participants was the importance of collaboration among farmers. Initially, many participants made individual decisions, focusing solely on their own strategies without considering cooperation with others. As the game progressed, only 40% realized that individual decision-making alone did not lead to increased income, as reflected by their inability to earn bonus points. However, after the game, around 80% of participants mentioned the necessity of working together to achieve better results.

Another key takeaway was the change in perception regarding gaming as a learning tool. Initially, many farmers were skeptical about participating in the session, as games are generally perceived as non-serious activities meant for children. However, after engaging in the gameplay, 93% of participants recognized its potential in simulating real-life choices and outcomes in a risk-free environment. The game allowed them to experiment with different decision-making strategies and immediately observe the consequences, something that is difficult to achieve in real life due to long-term uncertainties. This experience helped farmers appreciate the value of using simulations to test strategies before applying them in their actual farming practices.

The gameplay also revealed differences in engagement between young and older farmers. Younger farmers were more open and enthusiastic about the game, quickly adapting to its mechanics and actively exploring different strategies. Some even requested to replay the game, eager to refine their approach and maximize their outcomes

after understanding how it worked. This enthusiasm reflected their greater willingness to experiment and seek optimal solutions. In contrast, older farmers approached the game with a more measured and steady attitude. While they remained engaged and responded positively to the experience, they were generally less inclined to replay or modify their strategies significantly, preferring a more cautious and stable approach. This distinction highlighted the potential of interactive learning tools like serious games to engage younger farmers more effectively while still providing valuable insights for older farmers.

Discussions

Performance of RESORTES Game in Malang Context

Based on our finding, participants were able to communicate with one another and take part in open and lively conversation within the context of the RESORTES game. This session produced a set of expected qualitative and quantitative data regarding LU preferences based on farmer perspective. It demonstrates the successful implementation of the game in the context of Malang, apart from a minor adjustment made in regard to the game attributes, aligns with the findings of Andreotti et al. (2020), who also successfully applied the game in a Nicaraguan context to support collective scenario evaluation for sustainable landscape management. In many cases, adjustments in serious games are necessary to enhance the relevance of the game environment, allowing it to align more closely with real-world scenarios (Wouters et al., 2013). Such modifications have been shown to significantly improve participant engagement by fostering a deeper connection to the game setting, thus enriching the learning experience and promoting sustained interest (Deterding et al., 2011).

Participants' Choices in Game Sessions Close to Real-Life Decision

The similarity between participants' choices in both real-life and game sessions suggests that the design and setting of RESORTES may effectively simulate realistic dynamics, providing a reliable approximation of real-world decision-making within a game environment. This alignment between simulated and actual behavior has been observed in other game contexts as well; for instance, Levitt and List (2007) noted similar patterns, demonstrating how well-structured game designs can elicit behaviors comparable to those in real-world situations (Camerer & Hogarth, 1999). Realistic games can serve as a valuable research tool for studying participant decision-making and response patterns, as well as for investigating participant choices (Sari et al., 2024). Although the RESORTES game was first created to actively engage smallholders in the development of more sustainable agricultural landscapes and facilitate discussions and negotiations in community decision-making processes (Speelman et al., 2014), it has the potential to be utilised for other purposes as well. The RESORTES game may be used as a research method to understand the (actual) individuals' preferences and the

consistency of their decisions in complex social-ecological systems, as the decision-making during the game is similar to their actual practices.

Plot Distribution Preferences on Agricultural Decision-Making

In terms of game outcomes, a distinct preference for plot distribution was observed between locations. The choice of agricultural plots surrounding settlement areas may be influenced by soil fertility levels, as plots located in lower, concave slope positions tend to have higher fertility due to sediment deposits from upper slopes (Buraka et al., 2023). Farmers who preferred dispersed plots often prioritized reducing the risk of crop failure due to pest infestations over other factors, aligning with findings by Duong et al. (2019), who noted that pest and disease prevention is a primary concern for farmers in South Africa. Transportation efficiency for fertilizing was not a significant consideration for these participants, as agroforestry farmers in this region rarely fertilize their systems (Kusumawati et al., 2022; Saputra et al., 2022; Sari, 2023).

The Influence of Participant's Age on Land Use Choices and Its Consistency

The observed trend regarding the significant choice of land-use systems between farmer types was consistent with the findings of a study conducted in South Africa (Duong et al., 2019); however, it did not hold in the case of Brazil (Yigezu et al., 2018). This variation may be attributed to regional disparities, encompassing diverse socio-cultural conditions and environmental limitations, as well as economic and market-driven constraints. We surprisingly found that young farmers in Tulungrejo village exhibited a preference for crop-based LU schemes over tree-based ones. This result was supported by a recent study conducted by Sari et al. (2024), which found that the monocrop system gained prominence among young farmers due to its ability to generate high and rapid revenue. In this context, family size might influence these choices, as larger family sizes are frequently associated with higher living costs (UNFPA, 2018). In addition, economic pressures in newly formed young families (farmers) were reported to be higher, requiring higher living costs for food, education for children, etc. (Faturohman et al., 2023; Yodfiatfinda, 2020). Older farmers were more inclined to choose tree-based (agroforestry) systems to fulfil daily needs and lower labour and operational costs compared to crop-based systems.

The high consistency of agroforestry farmers in making tree-based LU decisions in this study validates previous research conducted by Sari et al. (2018), which demonstrated that the agroforestry system in this location exhibited a substantial utility value for farmers, ranging between 3.5 and 4.5 on a scale with a maximum score of 5. This implies that farmers value the presence of the agroforestry system by maintaining the system over time. It suggest the significant contribution of agroforestry systems to the local community. Sari, Kusumawati, et al. (2023) described that agroforestry systems in this area could provide the local communities with a strong sense of assurance regarding the long-term economic and ecological advantages. This assurance

appears to be a significant factor in farmers' consistent decision to retain their agricultural land without making any changes. Furthermore, the presence of agroforestry systems becomes more crucial for local communities since smallholder farmers rely on the system to meet their social and cultural necessities, such as conducting customary practices (Kusumawati et al., 2022) that remain significant in the present day. This can alleviate concerns regarding substantial uncertainty in agroforestry land cover changes, given the fact that the majority of agroforestry areas are located on privately owned land (Ollinaho & Kroger, 2021).

Impact of Societal Relations on Decision-Making in Game Outcomes

Based on the number of comments recorded during the game sessions, as presented in the results section, it appears that the level of active communication within a group is influenced by the strength of interpersonal connections among individuals. The proximity among participants can foster a collaborative environment, enabling seamless communication (Sanudin et al., 2023), as seen by the number of comments that appeared on a forum. The significant degree of engagement and dynamism exhibited by farmers, particularly in group 1, can be attributed to the strong interpersonal connections among participants, resulting in enjoyable circumstances that led to numerous comments and inputs being provided during each round of the game. An enjoyable environment in the forum could trigger farmers to discuss more freely and openly (Dupraz et al., 2009). This situation will enhance the synchronization among players and foster a sense of trust and reliance on one another. In this stage, collaborative effort to attain more points through group coordination could be tangible, as we observed a positive association between the relatedness index and additional points generated from collaboration effort (Figure 8(b)).

A low relatedness index might hinder the establishment of trust and impede the intended coordination, as reported by (Qurniati et al., 2017). The unsmooth coordination resulted in low total points collected without any additional points, which could be earned from coordination effort (Appendix 4). Based on that, fostering positive connections among farmers is a crucial element in building a robust social network, which facilitates the development of trust among them (Nyang'au et al., 2021). Particularly, when targeting participatory collaboration between smallholder farmers and other stakeholders to achieve more sustainable landscape management. The pattern of farmer preferences on LU selection, including the influenced social aspects behind decision making in the context of collective decision making, could benefit as a basic information for governments who want to involve the community in designing agroecosystems in a broader context of complex socio-environmental issues.

Recommendations

Based on participant feedback and researcher observations, several improvements can enhance the effectiveness and engagement of RESORTES in future applications. These

recommendations focus on increasing the game's relevance, appeal, and data collection efficiency. From the players' perspective, differentiating commodities under the land-use typology would make the game more engaging and reflective of real-world agricultural decisions. By allowing players to manage multiple crops with varying risks and returns, the game can better simulate the complex trade-offs farmers face. Furthermore, adding climate change and pandemic-related shocks as variables in the game would increase its relevance to current challenges. Farmers are increasingly facing unpredictable weather patterns and market disruptions, so incorporating these elements would allow them to explore adaptive strategies within a controlled simulation environment.

From the researchers' perspective, adjustments in terminology and methodology can improve participation and data accuracy. Given the perception that games are for children, using the term "simulation" instead of "game" may make farmers more receptive to participating, as it aligns better with their expectations of a serious learning tool. Additionally, audio recording should be mandatory, as some important details were missed during direct observation. This would provide a more comprehensive dataset for analyzing farmer interactions and decision-making processes. Lastly, facilitators should be proficient in the local language, as most participants naturally communicated in their native dialect during gameplay. Ensuring that facilitators understand and speak the local language would improve participant engagement, game facilitation, and data collection accuracy.

Conclusions

In conclusion, the 'serious game' simulation provided valuable insights into farmers' actual LU practices, revealing a strong preference for tree-based systems across both villages. The proximity of plots to residential areas emerged as a key determinant factor, emphasizing the significance of accessibility and efficiency in time and labor. Moreover, the findings underscored farmers' persistence in managing existing LU systems, particularly evident among tree-based systems, while crop farmers displayed comparatively less persistence. Importantly, communication, relatedness, and leadership emerged as influential factors shaping farmers' decision-making processes, highlighting the importance of social dynamics within agricultural communities.

As one of the potential sustainable LU practices, agroforestry is expected to minimize the trade-off between production and various ecological services. With the high level of farmer interest and consistency in choosing tree-based systems, the existence of agroforestry systems in the agricultural landscape may be long-lasting, contributing to a healthy and sustainable agricultural landscape. To further optimize these benefits, participatory-based planning needs to be explored further, especially in various different contexts, as the pattern of farmer preferences on LU selection, including the influenced social aspects behind decision-making, may vary. By gaining this understanding, future LU planning can be carried out appropriately, particularly on a community-based basis that prioritizes a bottom-up approach. An innovative

participatory approach with serious games can be optimized for guiding future agricultural landscape planning processes. Integration of these insights into agricultural policies and interventions can enhance the sustainability and resilience of rural landscapes while aligning with farmers' preferences and practices.

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Author Contributions

R.R.S., D.D.S. and E.S. conceived and designed the research; R.R.S., D.D.S., R.M.I., and E.P.S. performed data collections and game implementation; R.R.S., D.D.S., R.M.I., and E.P.S. analyzed the data. R.R.S. wrote the first draft of paper; all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Declaration of Conflicting Interest

The authors declare no conflict of interest.

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Ethical Statement

Ethical Approval

Our institution does not require ethical approval for reporting individual cases or case series.

Informed Consent

Informed consent was obtained verbally before participation.

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Data Availability Statement

All data will be made available based on the request.

Supplemental Material

Supplemental material for this article is available online.

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Erika N. Speelman is an associate professor of Spatial gaming and Simulation at Wageningen University. Erika is a tropical agronomist by training with a strong inter- and trans-disciplinary focus. Her work centers largely around contested tropical landscapes. She has been developing and implementing gaming and simulation tools to explore and facilitate (social) learning about the complex dynamics in social-ecological systems for nearly 20 years.